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Introduction to Computing for the Social Sciences

Session 3 – Data Structures

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Winter Semester 2026-2027

Course Outline

- *Session 1: Introduction*
- *Session 2: Information Coding*
- **Session 3: Data Structures**
- Session 4: Software
- Session 5: Algorithms
- Session 6: Recursion
- Session 7: Sorting Algorithms
- Session 8: Time Complexity
- Session 9: Theoretical Computer Science
- Session 10: Machine Learning
- Session 11: Databases
- Session 12: Parallel Programming
- Session 13: Social Science Applications
- Session 14: Exam Review

Outline

- Arrays
- Linked Lists
- Binary Search Trees
- Hash Tables

Arrays

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- **Arrays**
- Linked Lists
- Binary Search Trees
- Hash Tables

Two million respondents, one duplicate

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MOTIVATING EXAMPLE

- Your panel has **2 million** responses and you need to find the duplicates.
- Naive approach: compare every response with every other
- Takes **2×10^{12} comparisons**, hours to days on a laptop
- One pass with the right data structure does it in **seconds**.

$$2 * 10^{12}$$

comparisons, comparing every pair

$$2 * 10^{6}$$

operations, with a hash set

Why is “have I seen this before?” is nearly free in one structure and hopeless in another?

Six thousand rows in only 13 steps

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MOTIVATING EXAMPLE

A codebook has **6,000** entries and your script looks up a code for each of **40 million** observations. Checking the entries one at a time costs **6,000 comparisons** per lookup. If the codebook is sorted, it costs **13**.

6,000
comparisons, scanning the list

13
comparisons with binary search

460×
less work but the list must be sorted

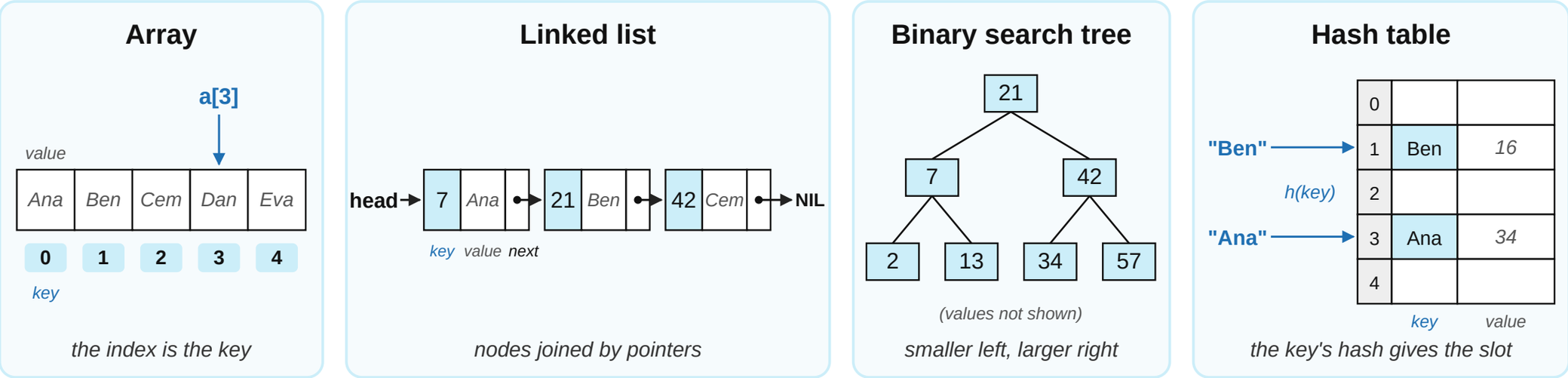
Why is this much faster and what is the problem if the codebook changes every day?

Data Structures

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Definition

Data types together with **operations** defined on these data that enable and realize access to and management of these data.



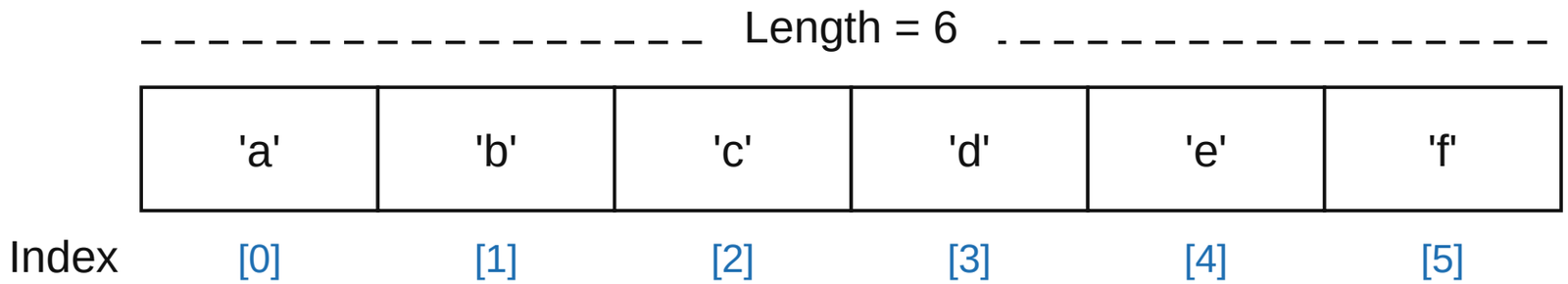
- We focus on data structures with particular relevance for information storage
- A search is not just to check if a **key** is there, also to retrieve **value** stored with it
- Diagrams often show only the keys; the values are implicit

Arrays

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Definition

Systematic arrangement of similar objects, usually in rows and columns.



- We address/access elements using the **index**
- in most programming languages indices range from *0* to *n-1* for an array of length *n*
- in MATLAB, R or **in pseudo-code we use indices *1* to *n***

Arrays: advantages and disadvantages

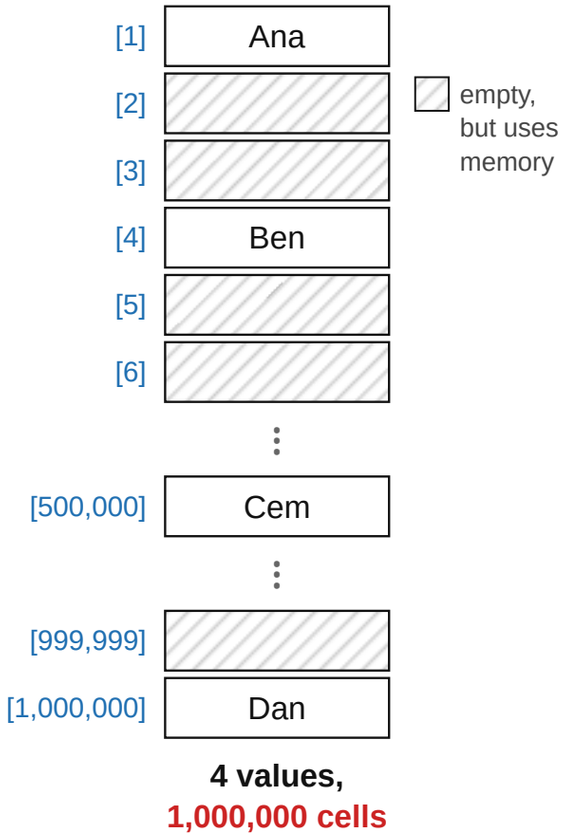
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Advantages

- Very fast access: if you know the index you only need one operation
- Standard structure implemented in most languages

Disadvantages

- Memory requirement depends on range of index values
  - If you have 4 entries with indices between 1 and 1,000,000, you need 1,000,000 cells in memory
- If just filling linearly, adding or removing in the middle leaves gaps or requires moving the rest of the content
- **What if you want indices that are not integers?**



Linked Lists and Dictionaries

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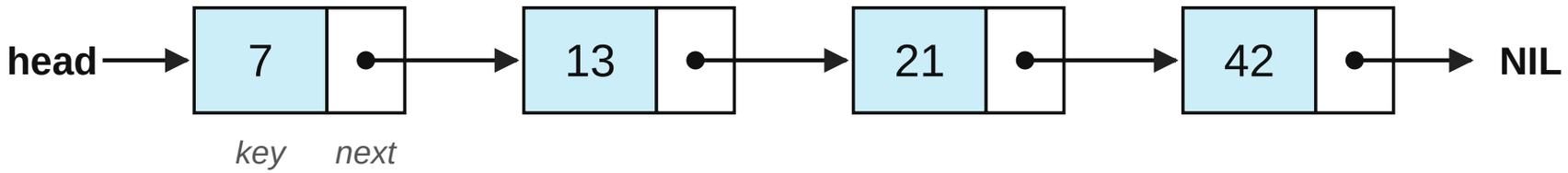
- Arrays
- **Linked Lists**
- Binary Search Trees
- Hash Tables

Linked Lists

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Definition

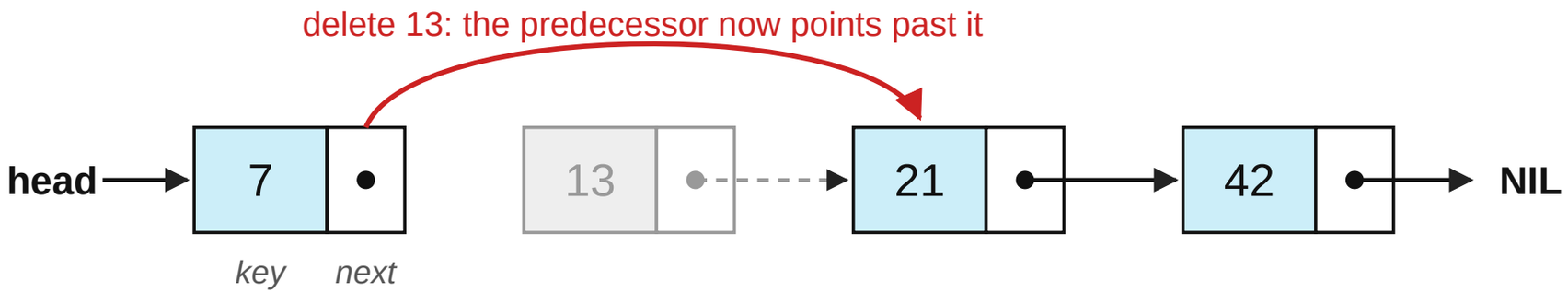
Linear collection of data elements, called nodes, each pointing to the next node by means of a *pointer*.



- Each node holds a **key** (and its data) and a pointer **next** to the following node
- The list is reached through a pointer **head** to the first node; the last node points to NIL
- Keys could be anything that can be checked for equality, not necessarily integers. E.g. strings, other pointers.

Linked Lists: advantages and disadvantages

Deleting a node only changes the pointer of its predecessor.



Advantages

- easy to edit: only adjacent pointers change
- memory grows with the number of nodes

Disadvantages

- hard to index, slow to iterate through
- extra memory for the pointers

Binary Search Trees

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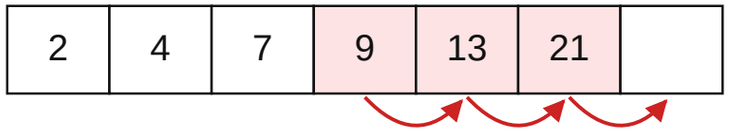
- Arrays
- Linked Lists
- **Binary Search Trees**
- Hash Tables

The question of editing vs search

- Linked lists have efficient editing but very slow (linear) search
- Arrays have fast search but inefficient editing
- Can we have some kind of search that is faster than linear but allows editing without moving all the data?
- When lists are sorted but we cannot directly index: **binary search**

Remember: we don’t search just for keys, we also retrieve values associated with them

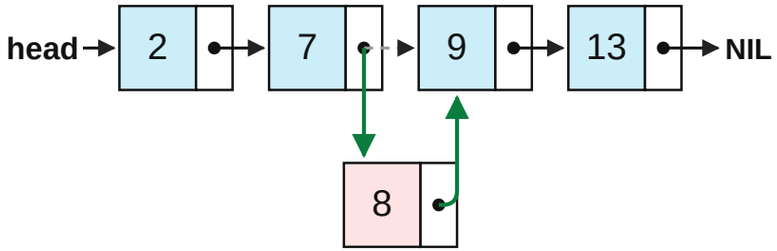
Array, sorted



✓ search: jump to any index by position

✗ insert 8 in position 4: shift 9, 13, 21

Linked list, sorted



✓ insert 8: change two pointers

✗ search: follow every pointer from head

Wanted: fast search and cheap editing

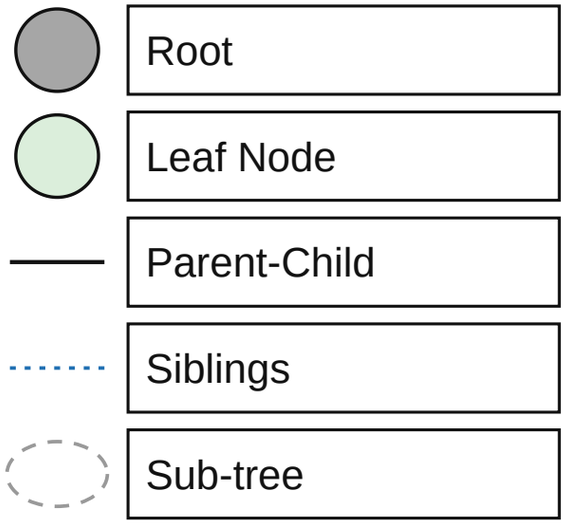
next: binary search, then binary search trees

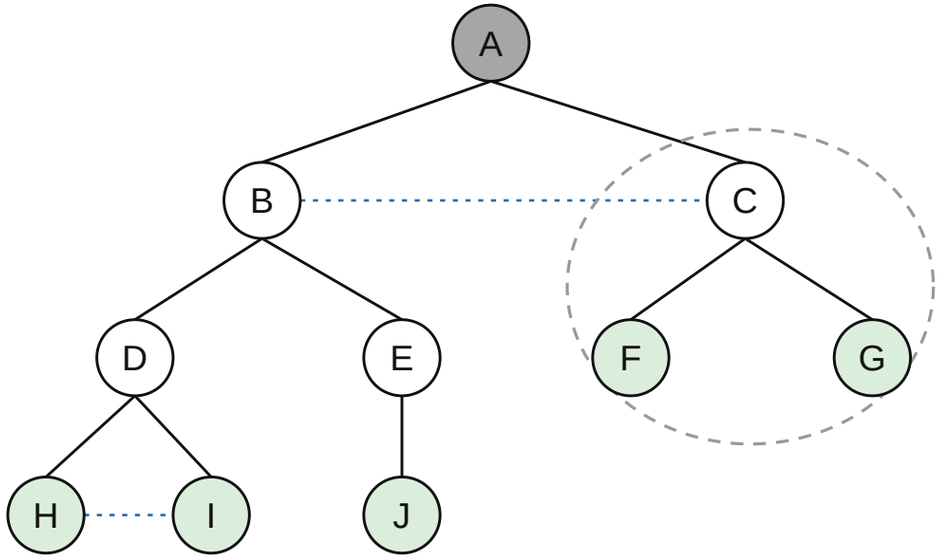
Binary Trees

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Definition: Binary Tree

A binary tree is a tree data structure that is either empty or it consists of a node (or root) S that has at most two children, which are referred to as the left child L and the right child R, both of which are also binary trees.





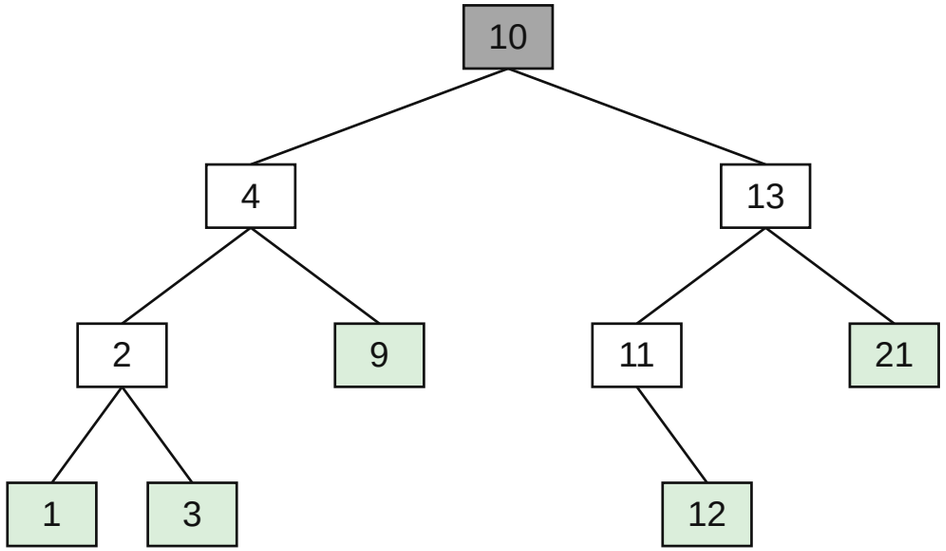
Binary Search Trees

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Definition: Binary Search Tree

A binary search tree is a binary tree with comparable keys where the key in the root has to be

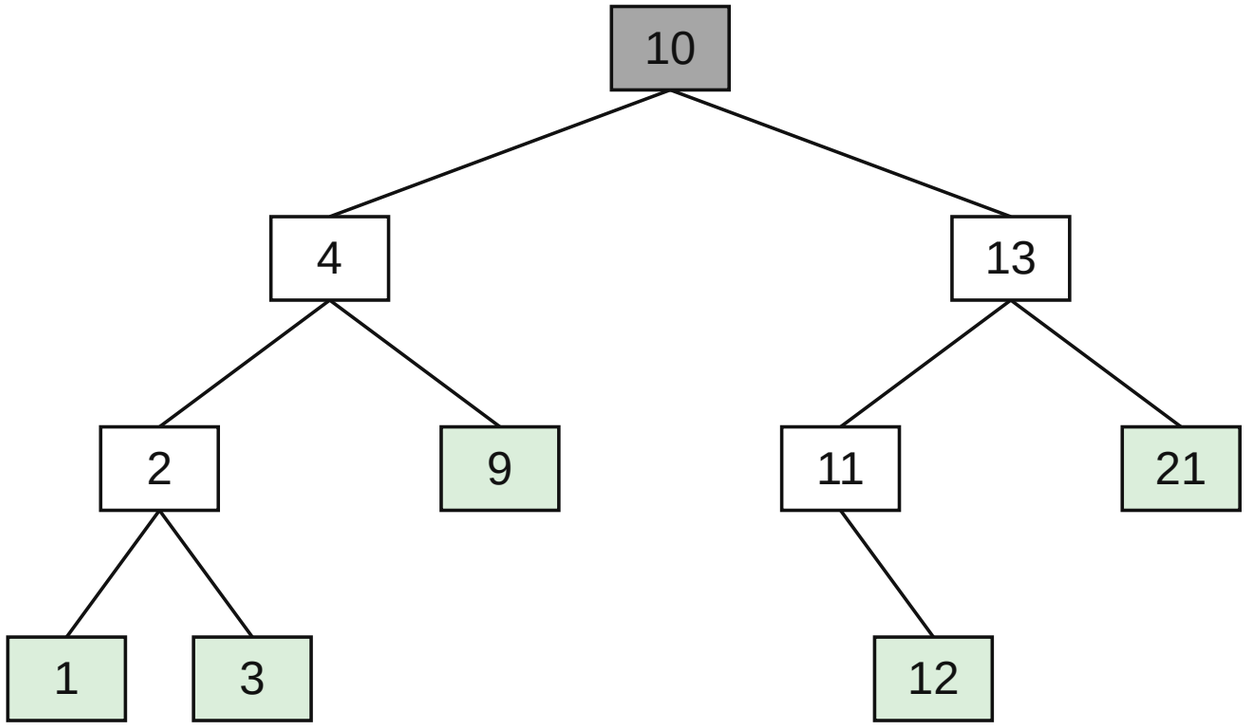
- **larger than or equal to** all keys in the left sub-tree, and
- **strictly smaller** than all keys in the right sub-tree.



- Remember that every element in the tree has a key but also contains data
- Here we assume that no duplicate keys are allowed

Binary Search Trees

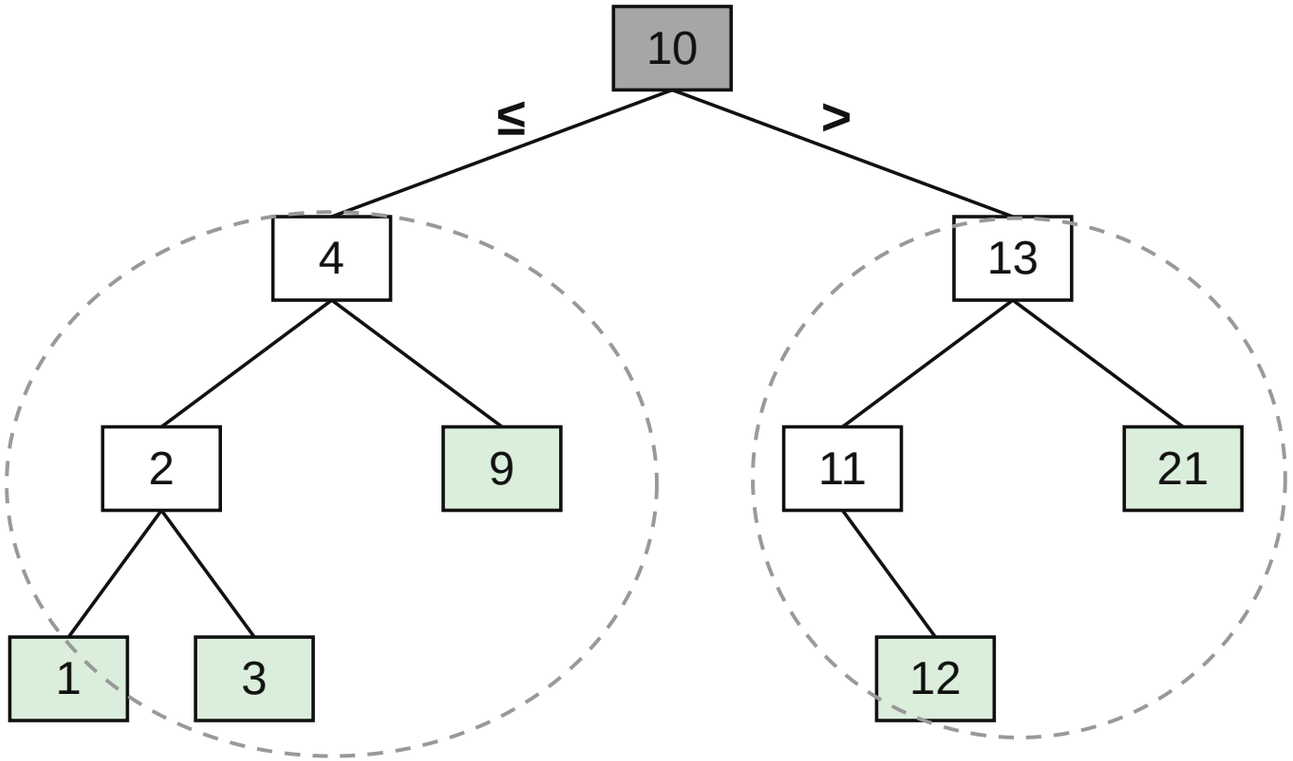
The tree we will use as the running example: ten keys



Binary Search Trees

the root has to be **larger or equal** to all keys in the **left sub-tree** and …

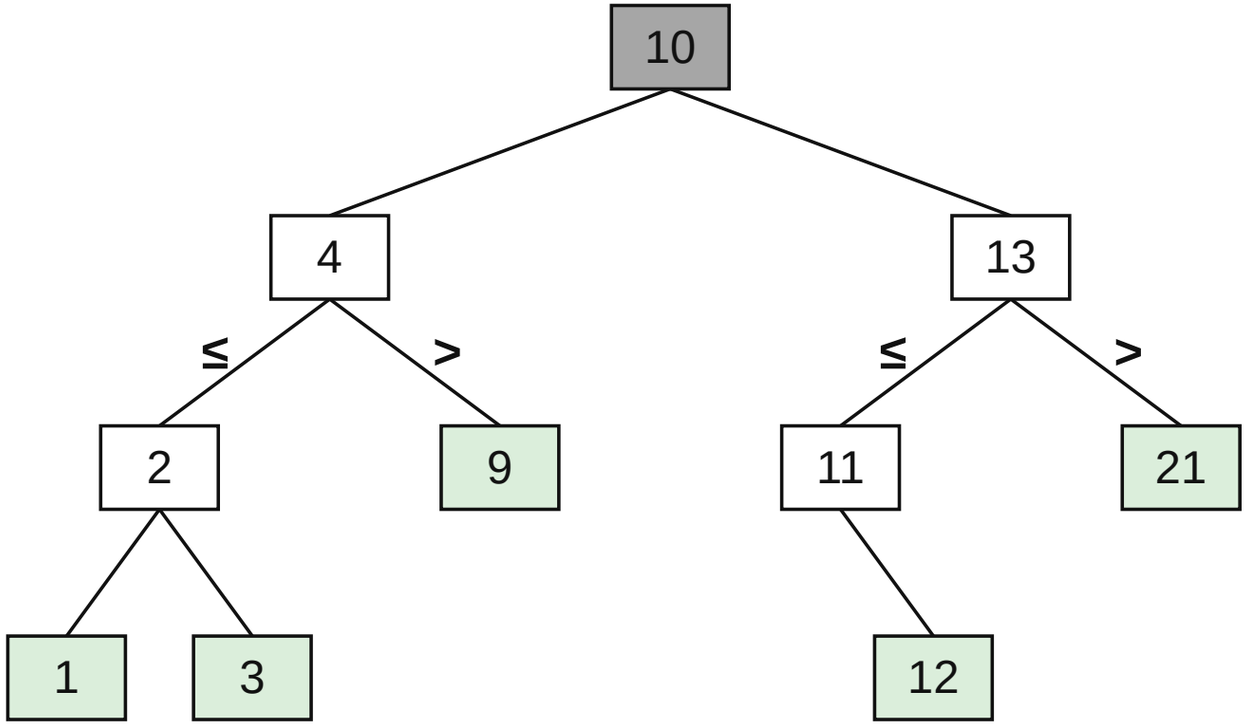
… **strictly smaller** than all keys in the **right sub-tree**



Binary Search Trees

the root has to be **larger or equal** to all keys in the **left sub-tree** and …

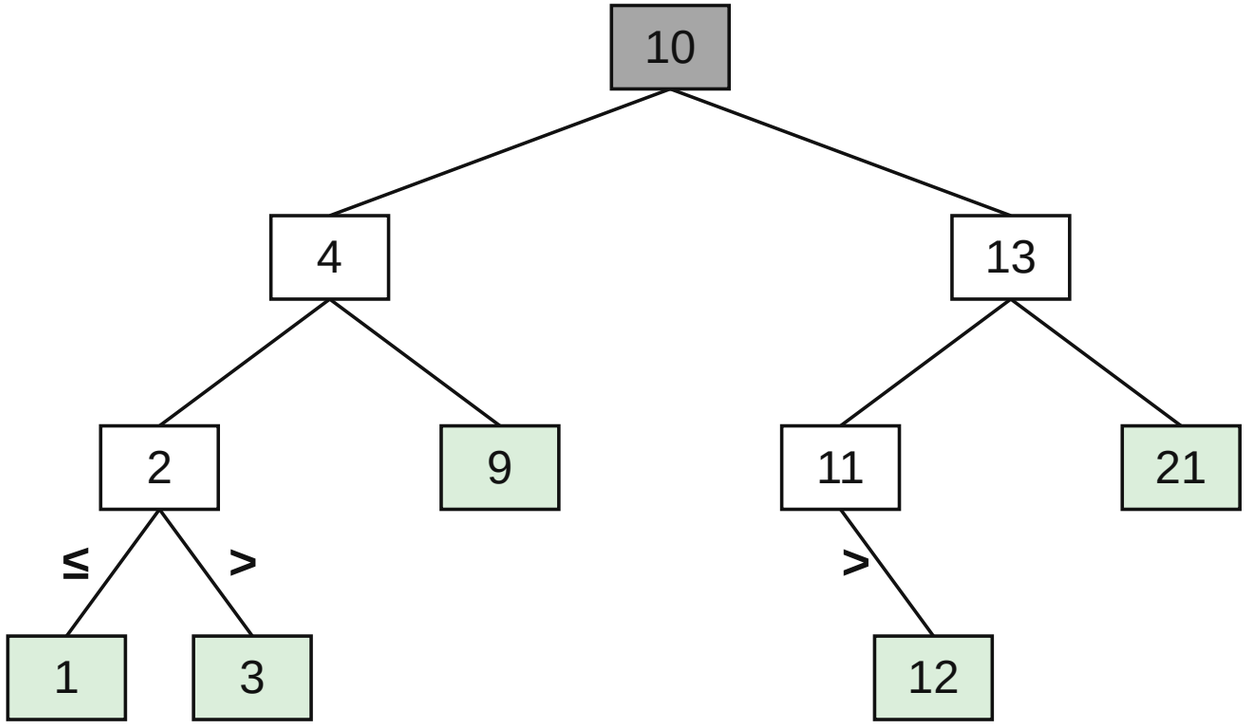
… **strictly smaller** than all keys in the **right sub-tree**



Binary Search Trees

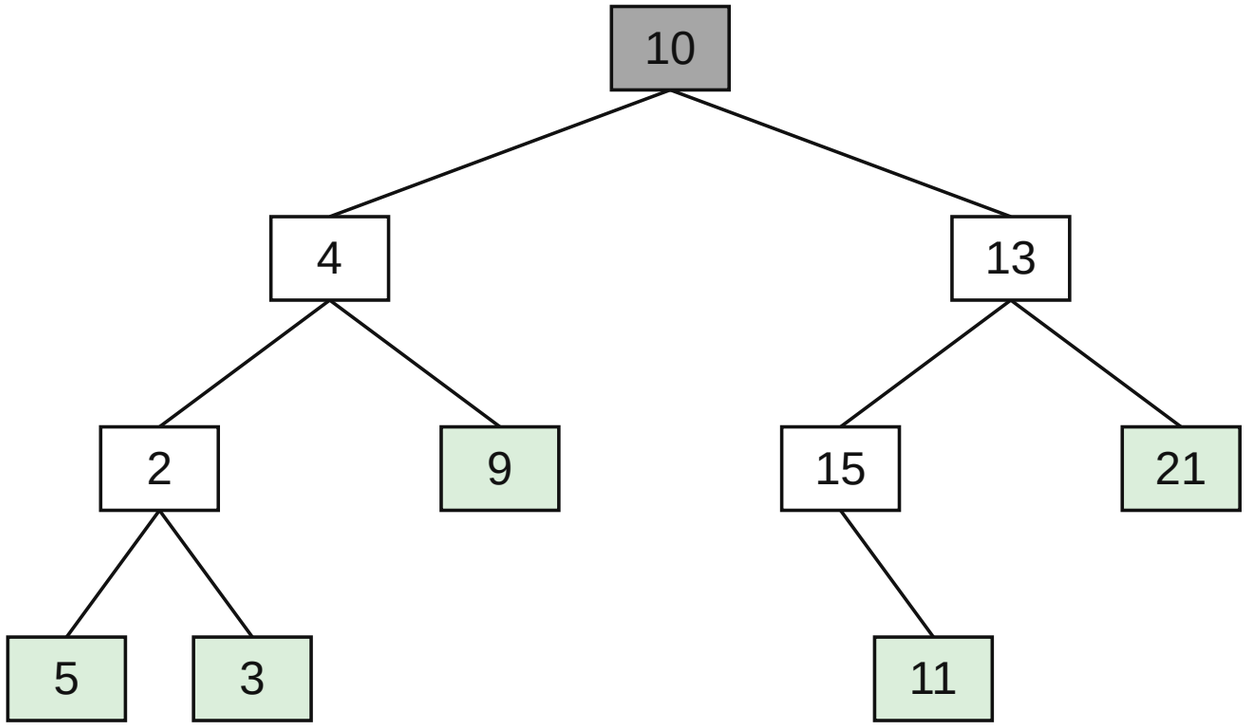
the root has to be **larger or equal** to all keys in the **left sub-tree** and …

… **strictly smaller** than all keys in the **right sub-tree**



Binary Search Trees

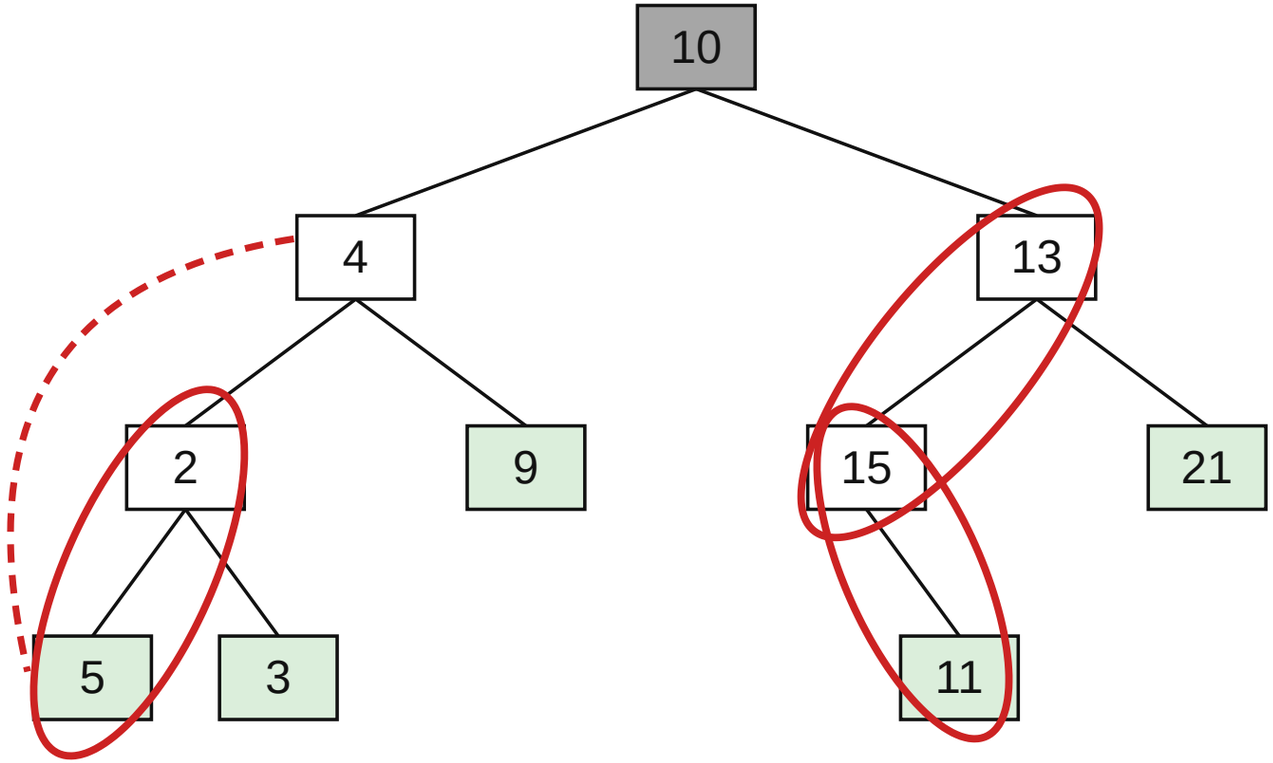
Is this a binary search tree?



Binary Search Trees

Is this a binary search tree? **No!**

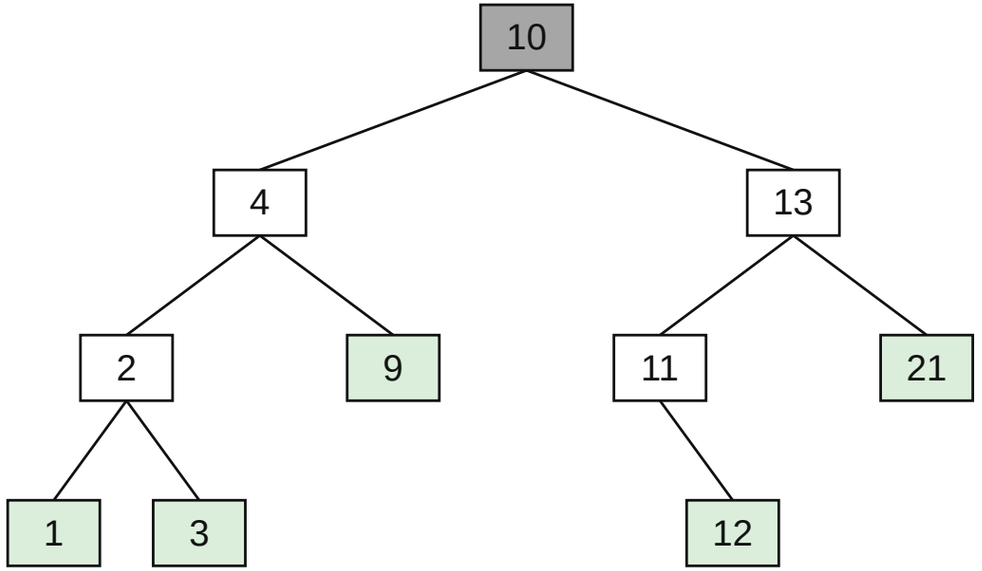
15 is in the left sub-tree of 13 but larger than 13; 5 is in the left sub-trees of 2 and 4 but larger than both



Binary Search Trees

Searching for a given key:

- compare the key you are looking for with the root
  - depending on whether it is smaller (larger) continue searching in left (right) sub-tree
  - return the value or *key not found*



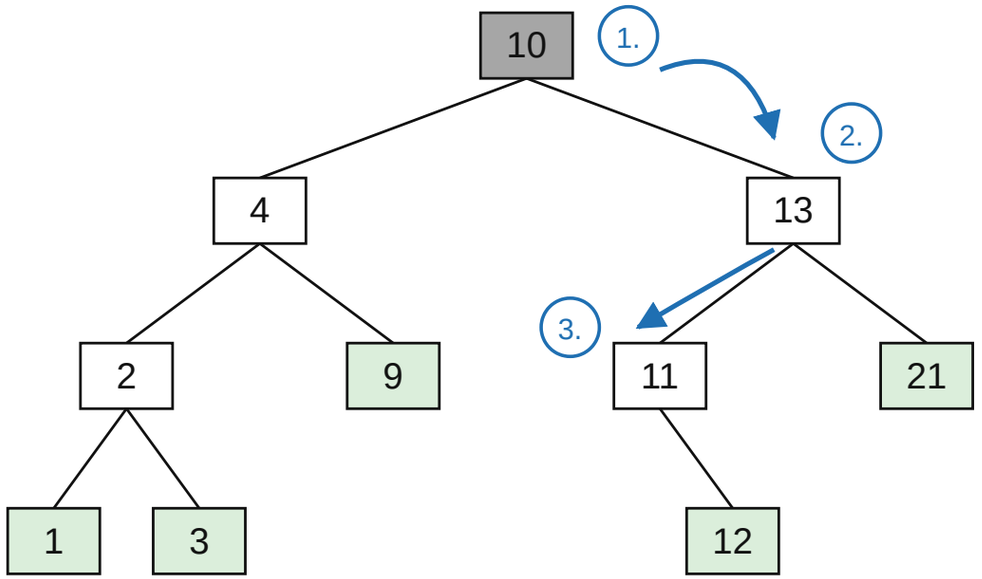
Binary Search Trees

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Searching for a given key:

- compare the key you are looking for with the root
  - depending on whether it is smaller (larger) continue searching in left (right) sub-tree
  - return the value or *key not found*
- Remember the Binary Search Algorithm

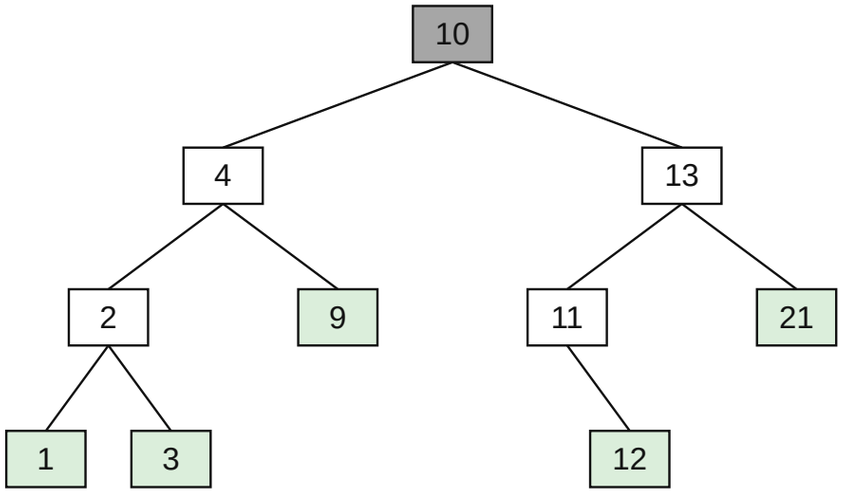
Search for 11:



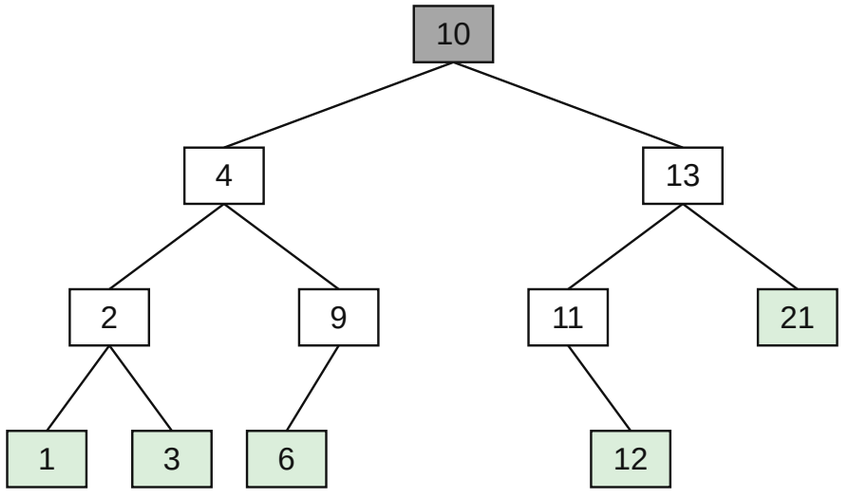
1. $11 > 10$ (root) → search on the right
2. $11 < 13$ → search on the left
3. $11 = 11$ → **Found 11**

Binary Search Trees

- **Adding** a new key at the correct position
  - assumption: key to be added is not yet in the tree
    - simply find appropriate empty slot and add the key as a “leaf”
    - only way to add keys and preserve binary search tree structure

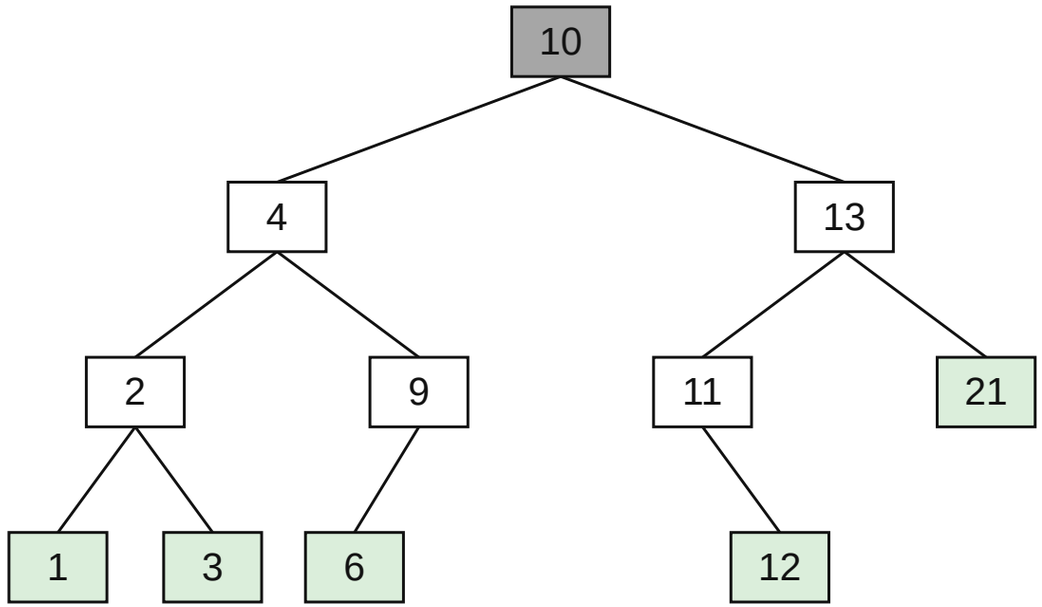


Add key 6:



Binary Search Trees

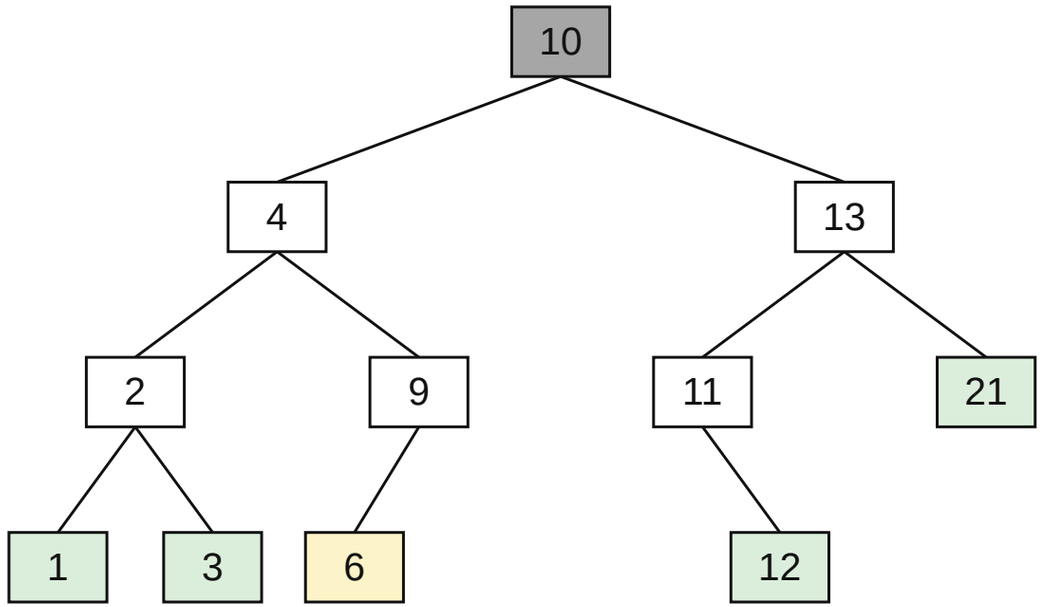
- **Delete** specific key while preserving a binary search tree structure
  - three cases: the node has no children (a "leaf" node), one child, or two children



Binary Search Trees

- **Delete** specific key while preserving a binary search tree structure
  - deleting a leaf node is simple:
    - search for key
    - in preceding node, remove reference to child

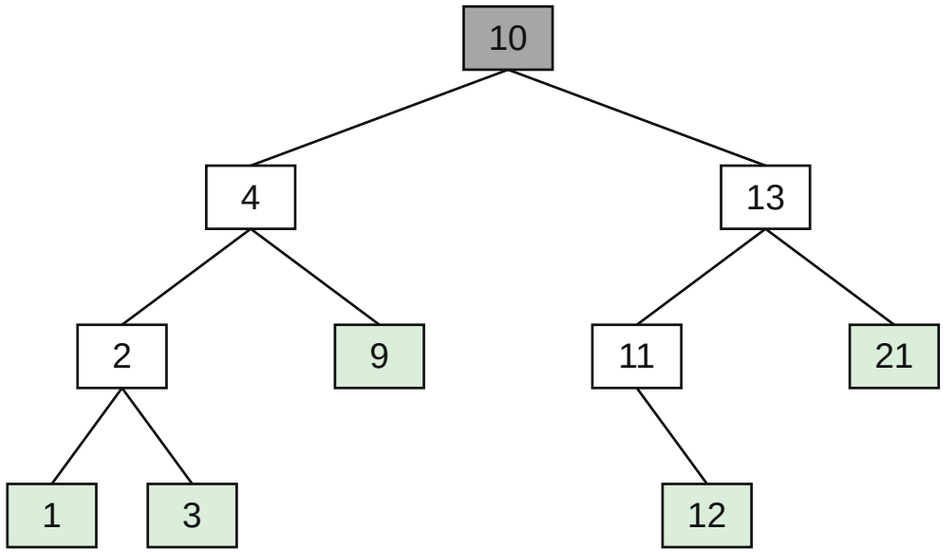
Delete key 6:



Binary Search Trees

- **Delete** specific key while preserving a binary search tree structure
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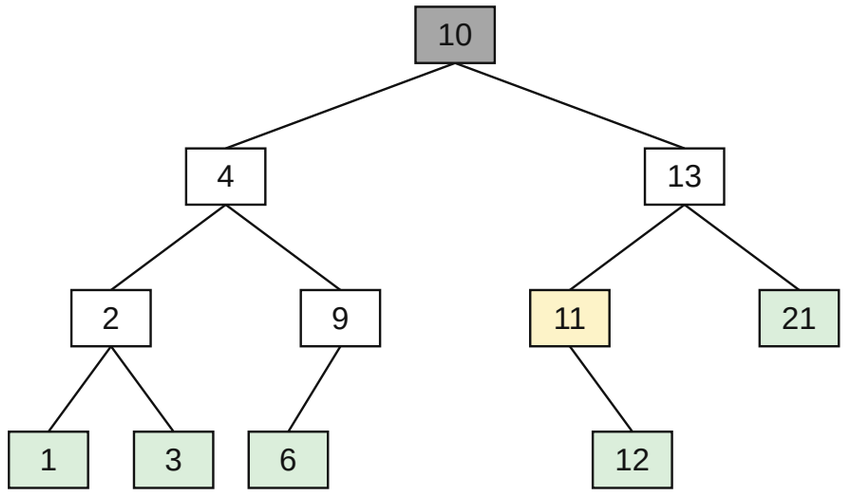
After deleting 6:

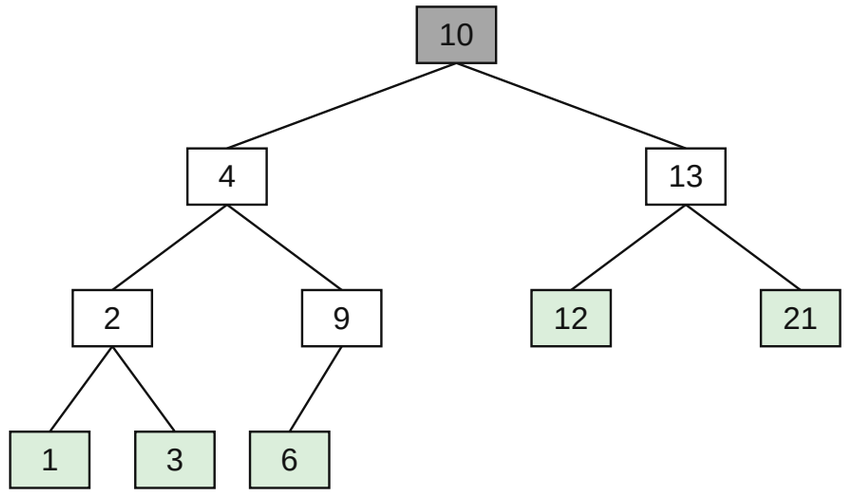


Binary Search Trees

- **Delete** specific key while preserving a binary search tree structure
  - deleting a node with one child is still relatively simple
    - replace it with its only child

Delete key 11:

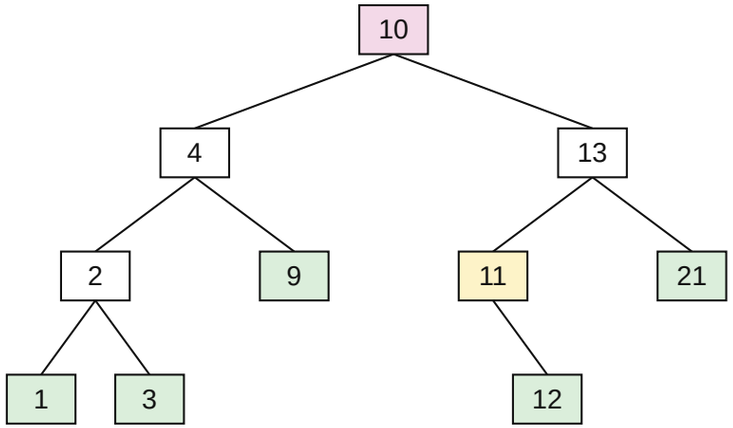




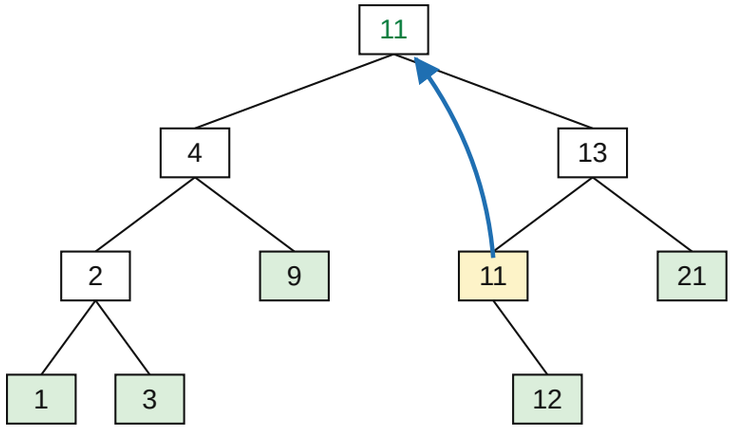
Binary Search Trees

- Deleting a node with two children is the most complicated case: replace it with the leftmost node of its right sub-tree (or the rightmost node of its left sub-tree), then restore the tree there, as in the one-child case

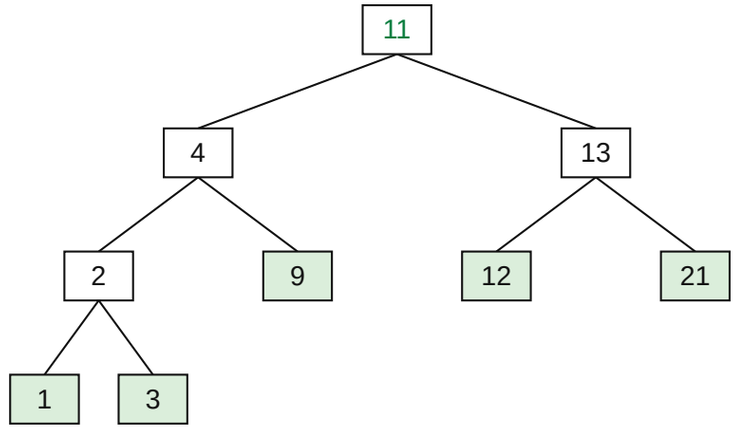
Delete the root 10:



1. the leftmost key of the right sub-tree is **11**



2. copy **11** into the root



3. remove the old **11**: its only child **12** takes its place

Binary Search Trees: advantages and applications

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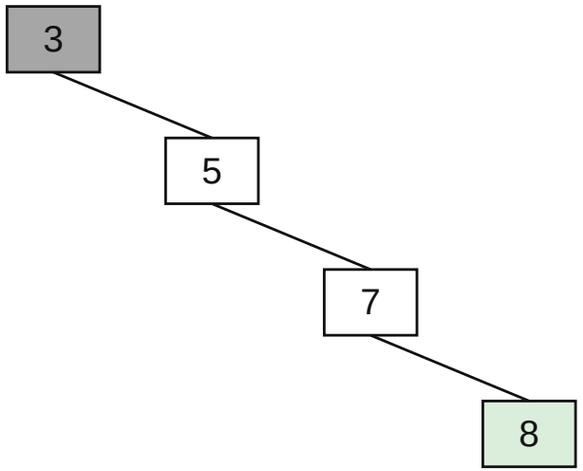
Advantages

- lookup for keys is much faster than in an unsorted array
  - same efficiency as sorted array with binary search
- adding/removing values is much faster than in a sorted array
  - just pointers are updated similarly to what is done in a linked list
  - in a sorted array all values to the “right” of the key have to be shifted

Practical applications

- used in look-up tables or dynamic sets
  - can vary in size but still efficient “housekeeping” of index keys
- it is ultimately most useful when paired up with data
  - fast look-up of data paired with each key, e.g. in database indices

Binary Search Trees: disadvantages



- Disadvantages of binary search trees
  - lookup takes a number of steps that depends on tree depth
  - if we add keys in an impractical order
    - many operations for deleting and adding
    - may arrive at a very unbalanced structure
    - in such a case no better than sequential search
    - need algorithms for balancing trees etc.

Hash Tables

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- Arrays
- Linked Lists
- Binary Search Trees
- **Hash Tables**

Hashing

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- Can we have very fast lookup (about one operation per access) but not have the index range problem of arrays?
- Can we use other indices besides integers?

- Examples:
  - student ID numbers
    - 6-digit number, i.e. 1,000,000 possible values
    - BUT only 10,000 are used at the same time
    - can map all numbers onto 10,000 slots, no need for 1,000,000
  - books in the library
    - book titles have maybe 100 characters, that means 30^{100} possible titles
    - BUT the library has “only” about ~2 million books

Hash Functions

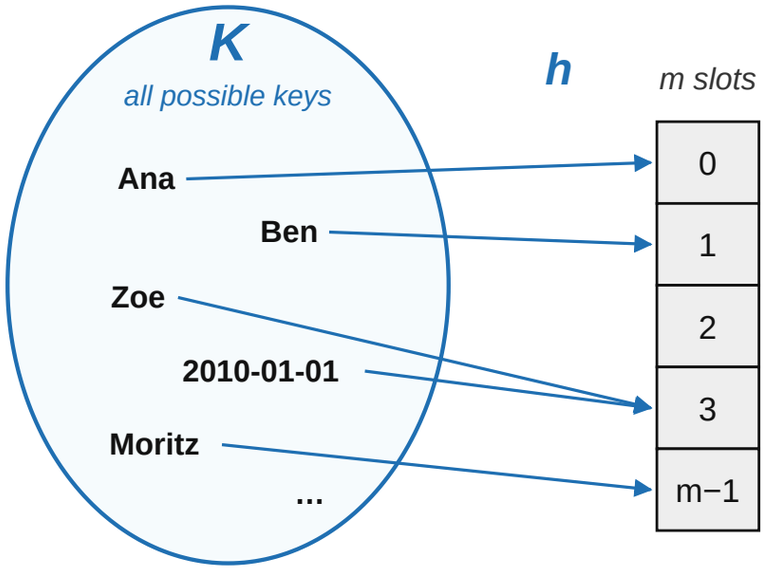
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Definition: hash function

A hash function $h : K \to \{0, 1, \ldots, m-1\}$ assigns to any given key k an index $0 \leq h(k) \leq m-1$.

Requirements for $h(k)$:

- $|K| \gg m$, i.e. many more possible keys k than values $h(k)$ maps to
- fast to compute, deterministic
- balanced distribution to available indices to avoid collisions
- (dynamically) adjustable to the number of free slots m



hash functions are usually based on natural numbers (with 0), i.e., usually direct calculation of the hash value of a given key (e.g., as a character string)

Hash Tables

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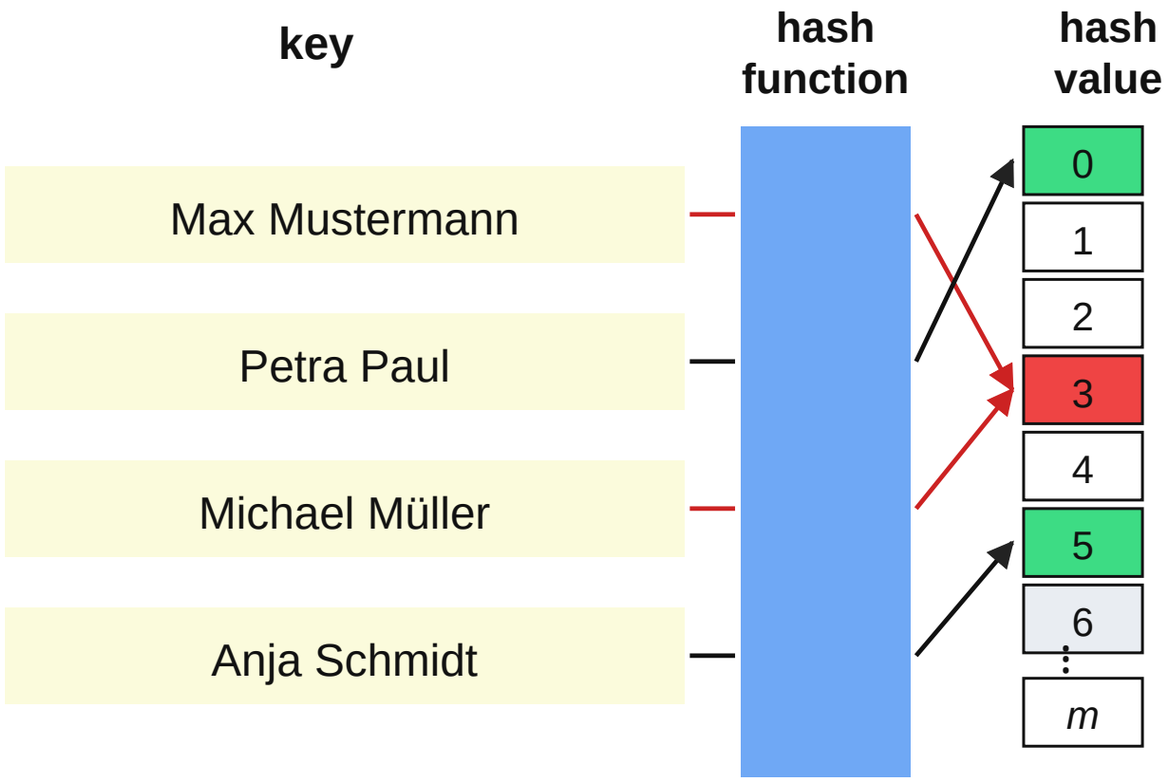
Operations on hash tables: each item says how to do one operation for a key k

- searching for an entry:
  - calculate hash function $h(k)$ and read position $h(k)$
- adding an entry:
  - calculate hash function $h(k)$ and write at position $h(k)$ (what if the position is used?)
- deleting an entry:
  - calculate hash function $h(k)$ and empty position $h(k)$

Time required for operations: just one step in the best case

Hash Collisions

Hash collisions happen when two keys are mapped to the same position by the hash function



Hash Tables

- Modulo operation
  - automatically yields values in the range 0 to $m - 1$
    - ideal hash function since it homogeneously distributes values
  - important to select the “right” m to avoid hash collisions
    - best choice are prime numbers
    - worst choice are powers of the base of the number system used
- Example with problems:
  - dates represented as numbers

key k	$k \bmod 97$	$k \bmod 100$
20100101	52	1
20100227	81	**27**
20100427	87	**27**
20110527	**2**	**27**
20090811	74	**11**
20101224	11	**24**
20111111	**4**	**11**
20120229	**4**	29
20091224	**2**	**24**

Bold: collisions. $\bmod 100$ keeps only the day

Hash Collisions - Separate Chaining

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Separate chaining is one solution to collisions: every slot holds a linked list of all the keys hashed to it.

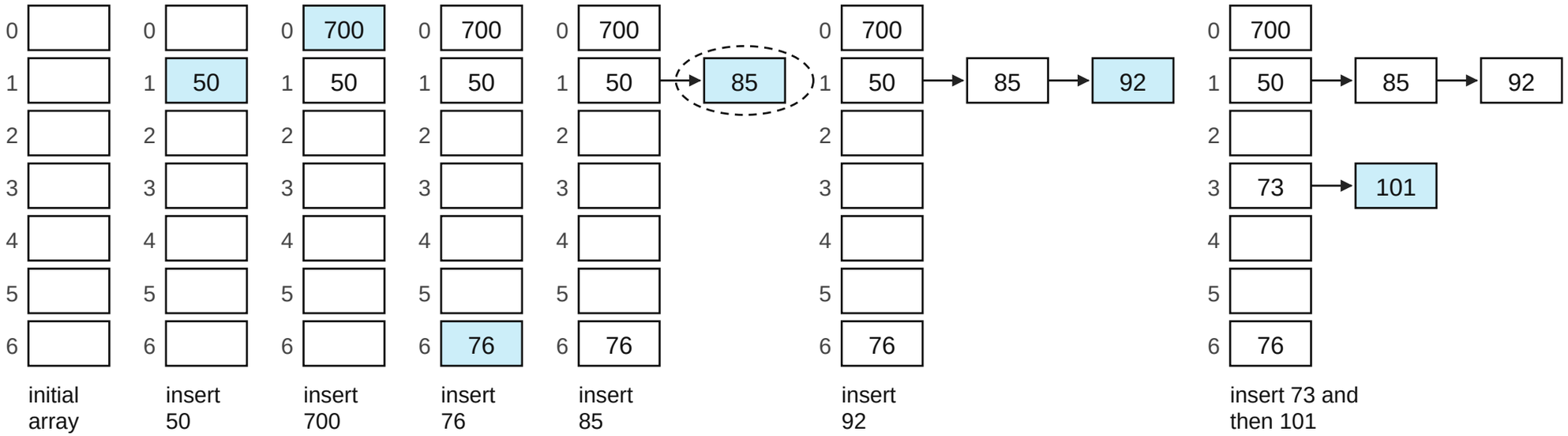
Operations on hash tables with separate chaining:

- **searching** for an entry: calculate $h(k)$, then search through the linked list of that slot from front to end
- **adding** an entry: calculate $h(k)$, search the list, and append the new element if it is not there
- **deleting** an entry: calculate $h(k)$, search the list, and remove the element from it

Time required for operations: just one step in the best case, n steps in the worst case

Hash Collisions - Separate Chaining

Dealing with hash collisions through **separate chaining**: each entry of the hash table is a linked list, and colliding entries are appended to it

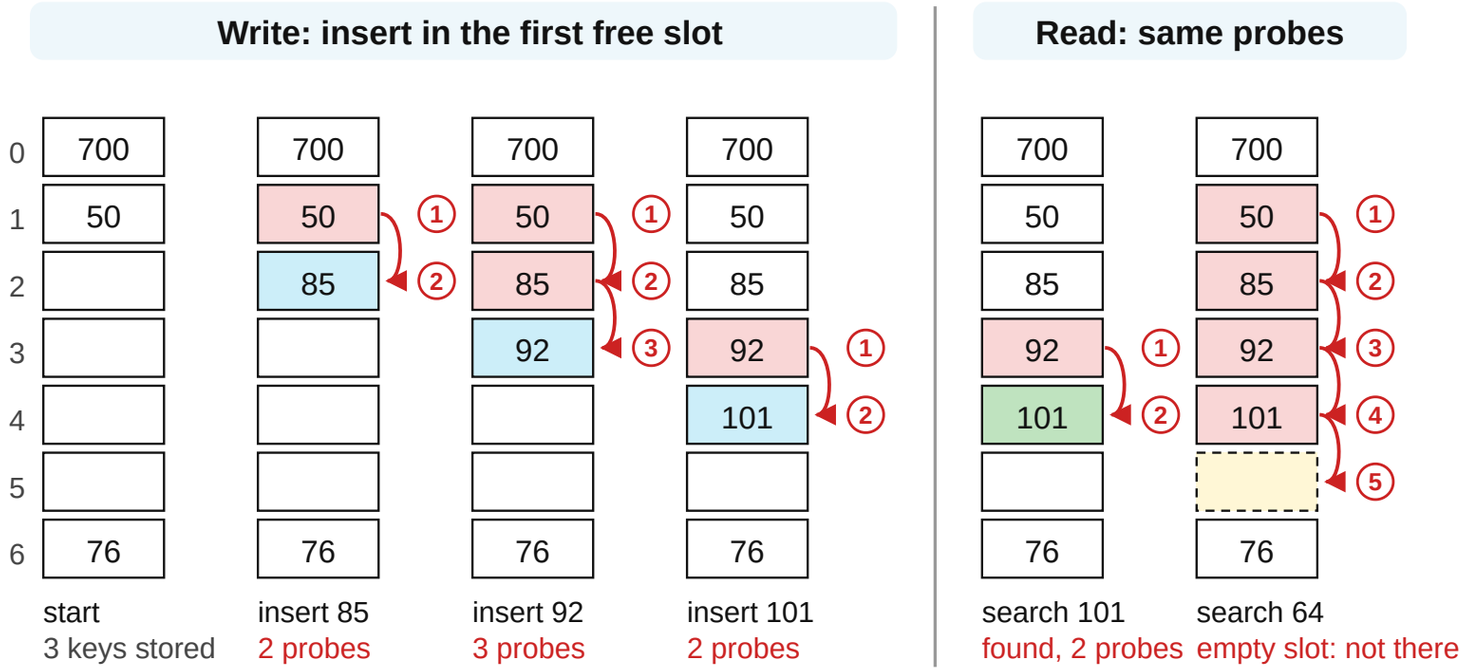


Example: $m = 7, h(k) = k \bmod 7$

Hash Collisions - Linear Probing

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Open addressing stores every key in the table itself. **Linear probing**: if slot $h(k)$ is taken, try $h(k)+1$, $h(k)+2$, … (mod m). **Write** in the first free slot; **read** by following the same probes until the key or an empty slot

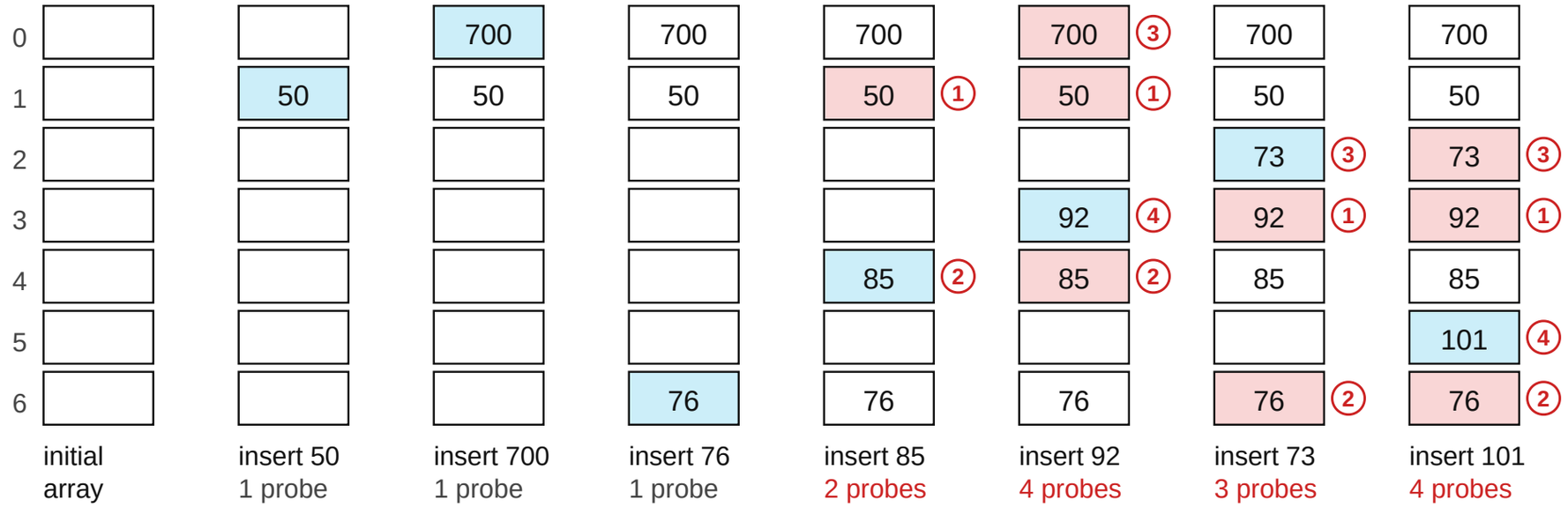


$m = 7$, $h(k) = k \bmod 7$; 700, 50 and 76 are already stored; circled numbers: order of probes

Hash Collisions - Random Probing

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Random probing: if slot $h(k)$ is taken, try $h(k) + r_1$, $h(k) + r_2$, ... (mod m), with a fixed pseudo-random sequence r (here 3, 6, 2, 5, 1, 4) that a search can repeat. After inserting 92 the keys are spread out, not in one run



Keys 50, 700, 76, 85, 92, 73, 101 inserted in this order: $m = 7$, $h(k) = k \bmod 7$. Circled numbers give the order of the probes

Hash Collisions - Probing: what to watch

- **Clustering**: with linear probing, occupied slots form runs, and every key that hashes into a run walks to its end and makes it longer. In the linear probing example, a new key hashing to any slot except 5 ends in slot 5
- **Random probing** avoids these runs, but its sequence must be reproducible (fixed, or computed from the key), or a search could not find the key again
- **Load**: both slow down sharply as the table fills. With n keys in m slots, keep n/m well below 1: Python's `dict` grows when it is two thirds full
- **Deletion**: emptying a slot would cut the probe sequence of the keys stored after it. Mark the slot as *deleted* instead: searches skip it, insertions may reuse it

Home exercise. In the linear probing example, delete 85 by marking its slot as *deleted*. Then search for 92 and for 101: which slots are probed? Then insert 64: where does it go? What would go wrong if slot 2 were simply emptied?

Dictionaries

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Definition

Abstract data type composed of a collection of (key, value) pairs, such that each possible key appears at most once in the collection.

- Generalized keys (not just indices):
  - characters, words, names etc.
- Characteristics
  - keys are not necessarily homogeneously distributed
    - using an array spanning the whole range of values would be a waste of memory
  - use list of keys to index the dictionary
    - if no further conditions applied: sequential search
    - worst case: have to search through ALL n entries

Dictionaries in Python

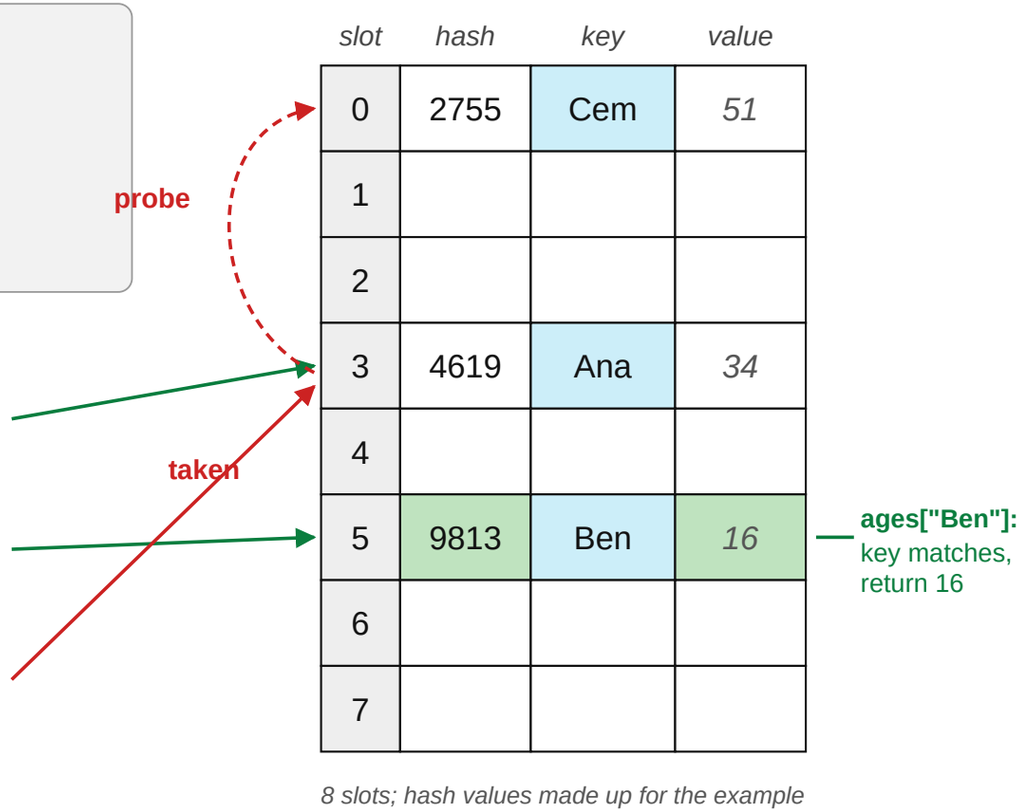
- Python uses **hash tables** to store dictionaries
  - **Collisions are solved with random probing:** when a slot is full, move to another slot in a pseudo-random sequence that is computed from the key, so a search can repeat it
  - Hashes start with 8 slots and are resized when ⅔ of entries are full

```
ages = {}
ages["Ana"] = 34
ages["Ben"] = 16
ages["Cem"] = 51
ages["Ben"]   → 16
```

hash("Ana") = 4619
4619 mod 8 = 3

hash("Ben") = 9813
9813 mod 8 = 5

hash("Cem") = 2755
2755 mod 8 = 3



8 slots; hash values made up for the example

https://stackoverflow.com/questions/327311/how-are-pythons-built-in-dictionaries-implemented/

Dataset pseudonymization

name	wave	age	vote
Ana Weber	1	34	yes
Ben Koch	1	27	no
Clara Roth	1	45	yes
Ana Weber	2	35	no

h(secret + name)
→

person	wave	age	vote
c3305900d9	1	34	yes
f61798a8f3	1	27	no
f109d00beb	1	45	yes
c3305900d9	2	35	no

- Replace the identifying column by its hash: the same name always gives the same code, so the two waves of Ana Weber can still be linked
- A hash cannot be turned back into the name, but it can be **guessed with a lot of effort**: hash every name in a phone book and compare
- Approach: hash the name together with a **secret** that only the research team knows
- Pseudonymized is not anonymous: age and answers together may still identify a person
- In the example, the code is the first 10 characters of SHA-256 of the secret followed by the name.

Brian Kernighan and AWK

- The AWK language is a specialized language for string processing: a secret tool of many data scientists
- key component of UNIX, co-created by Brian Kernighan at Bell Labs
- AWK: Alfred **A**ho, Peter **W**einberger and Brian **K**ernighan
- Extremely efficient thanks to the use of regular expressions (we will learn them in session 9) and hash tables
- The default array structure is an associative memory (dictionary) based on a hash table
- MAWK: faster implementation and more memory efficient



Brian Kernighan on hash tables: https://www.youtube.com/watch?v=qTZJLJ3Gm6Q

Summary

- A data structure combines data with operations; the right one makes the frequent operations cheap
- **Arrays**: one step to access by index, but a fixed index range and costly insertion in the middle
- **Linked lists**: cheap insertion and deletion, but a search walks the list
- **Binary search trees**: search, insert and delete in as many steps as the tree is deep
- **Hash tables**: about one step per lookup; collisions are handled by separate chaining or probing
- Python's `dict` is a hash table: use it for lookups by key, lists for ordered sequences

Next session: Software: programming paradigms, good coding practices, software engineering

Appendix: Binary Search

Procedure BINARY-SEARCH(*A*, *n*, *x*)
Inputs:
    A: a sorted array
    n: the number of elements in *A*
    x: the value to find
Result: an index *i* with *A*[*i*] = *x*, or NOT-FOUND if *x* is not in *A*
p ← 1; *r* ← *n*
While *p* ≤ *r*:
    q ← ⌊(*p* + *r*)/2⌋
    If *A*[*q*] = *x*
        then return *q*
    else if *A*[*q*] > *x*
        then *r* ← *q* - 1
    else *p* ← *q* + 1
return NOT-FOUND

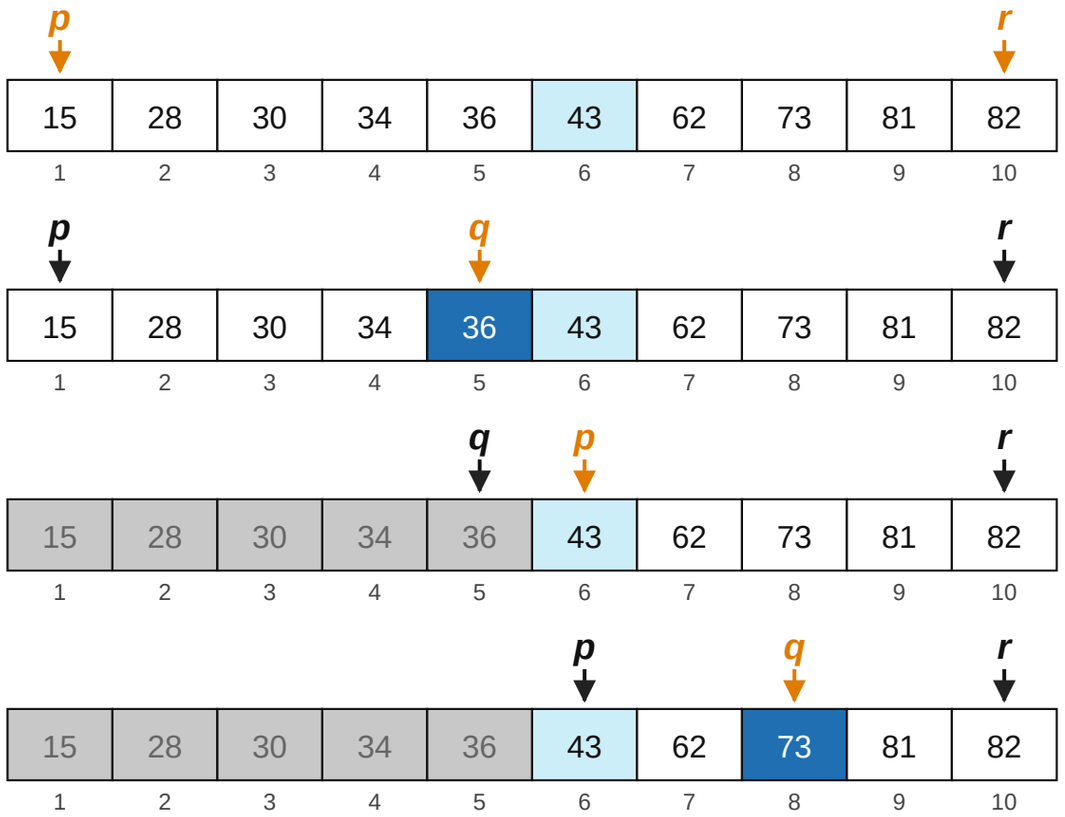
- In ordered array A of size n, search for the value x

See Cormen (2013), pp. 28–30, for more on this algorithm

Appendix: Binary Search

Procedure BINARY-SEARCH(A, n, x)
Inputs:
A: a sorted array
n: the number of elements in A
x: the value to find
Result: an index i with $A[i] = x$, or NOT-FOUND if x is not in A
$p \leftarrow 1$; $r \leftarrow n$
While $p \leq r$:
$q \leftarrow \lfloor (p + r)/2 \rfloor$
If $A[q] = x$
then return q
else if $A[q] > x$
then $r \leftarrow q - 1$
else $p \leftarrow q + 1$
return NOT-FOUND

Use binary search to **find 43**



Appendix: Binary Search

Procedure BINARY-SEARCH(A, n, x)
Inputs:
A: a sorted array
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Result: an index i with $A[i] = x$, or NOT-FOUND if x is not in A
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return NOT-FOUND

Use binary search to **find 43**

